

The Odd Symmetric Ideal of the K_6 Matching Arrangement and a Frobenius Descent Theorem

with an application to the integral Hodge conjecture for Fermat fourfolds (Degtyarev-Shimada Conjecture 1.2)

Rafael Amichis Luengo (Madrid), with the Constructor and Auditor (Claude) · Operación Glotón / The Sofa Theorem · 3 July 2026

Abstract. Let $S = F_3[s_0, \dots, s_5]$ and, for each of the fifteen perfect matchings J of the six indices, let $I_J = (s_a + s_b : \{a, b\} \in J)$ be the ideal of the 3-plane on which the variables pair with opposite signs. We prove two theorems. (A) The intersection of the fifteen leg ideals is the odd symmetric ideal: $\cap_J I_J = (e_1, e_3, e_5)$, a reduced complete intersection of dimension 3 and degree 15 whose components are exactly the fifteen planes; the proof is characteristic-free away from 2. (B) For every $v \geq 1$, with $Q = 3^v$ and $m_Q = (s_0^Q, \dots, s_5^Q)$ the Frobenius-power ideal, $\cap_J (I_J + m_Q) = (e_1, e_3, e_5) + m_Q$. The proof of (B) is an induction on v whose step is purely structural — three short lemmas resting on the freeness of S over its cube subring and on the additivity of the Frobenius in characteristic 3 — and whose base ($v = 1$) is a finite statement verified exactly by two independent computations, one primal and one through Gorenstein duality, with pencil arguments closing all but degrees 3–12. The step proves, more generally, that for any arrangement of linear-form ideals over F_p , saturation against Frobenius powers at the first level implies it at every level. Via a sealed reduction chain of the authors' campaign (cited, with sufficiency direction), Theorem B yields $\text{Tors}_3(\text{coker } \Delta_v) = 0$ for all v — the Degtyarev-Shimada Conjecture 1.2 in dimension 4, open since 2015. All verification code and logs are published with the campaign corpus.

1. Statement of results

Fix $S = F_3[s_0, \dots, s_5]$. A perfect matching J of $\{0, \dots, 5\}$ determines the codimension-3 linear ideal $I_J = (s_a + s_b : \{a, b\} \in J)$ and the 3-plane $V_J = V(I_J)$; there are fifteen matchings, forming one orbit of S_6 . Let e_k denote the elementary symmetric polynomials in the six variables.

Theorem A (Odd Symmetric). Over any field of characteristic $\neq 2$: $\cap_J I_J = (e_1, e_3, e_5)$. Moreover e_1, e_3, e_5 is a regular sequence; $S/(e_1, e_3, e_5)$ is a reduced complete intersection of dimension 3 and degree $1 \cdot 3 \cdot 5 = 15$, its components being exactly the fifteen planes V_J . No even elementary symmetric polynomial lies in any I_J .

Theorem B (Frobenius Descent). For every $v \geq 1$, writing $Q = 3^v$ and $m_Q = (s_0^Q, \dots, s_5^Q)$: $\cap_J (I_J + m_Q) = (e_1, e_3, e_5) + m_Q$.

Descent Principle (general form of the inductive step). Let $\{I_\alpha\}$ be any finite family of ideals of $F_p[s_1, \dots, s_n]$ generated by linear forms, and $E = \cap_\alpha I_\alpha$. If $\cap_\alpha (I_\alpha + m_p) = E + m_p$, then $\cap_\alpha (I_\alpha + m_{p^v}) = E + m_{p^v}$ for every $v \geq 1$.

The application (Section 5): Theorem B is the statement $(\square_{34})$ of the authors' campaign; through the campaign's sealed reduction chain — sufficiency established in the corpus and cited by address — it implies $\dim T(v) = P(3^v)$ at all v , the collapse of the rank chain, $\text{drop}(\Delta_v) = 0$, and hence $\text{Tors}_3(\text{coker } \Delta_v) = 0$ for all v : the saturation of the diagonal $\Sigma 1_J = \text{im}(\Delta)$ at every level of the Fermat tower, i.e. Degtyarev-Shimada Conjecture 1.2 in dimension 4.

2. Preliminaries

$F : S \rightarrow S$, $a \mapsto a^3$, is the Frobenius endomorphism; $F(S) = F_3[s_0^3, \dots, s_5^3]$. We use four standard facts, each with a two-line proof. (L2) Cubing is F_3 -linear on linear forms $((\sum a_i s_i)^3 = \sum a_i s_i^3, a_i^3 = a_i)$, hence for any frame — any F_3 -basis $y_1, \dots, y_6$ of the linear forms — one has $m_Q = (y_1^Q, \dots, y_6^Q)$, and $\varphi^Q \in m_Q$ for every form φ . (K) S is a free $F(S)$ -module with basis the 3^6 reduced monomials (all exponents ≤ 2) of any frame; every $t \in S$ has a unique digit expansion $t = \sum_\rho F(t_\rho)\rho$ per frame. (D) $t \in m_Q$ if and only if every digit $t_\rho \in m_{Q'}$, where $Q = 3Q'$. (G) $R_Q = S/m_Q$ is an Artinian complete intersection, hence Gorenstein; its annihilator map is an inclusion-reversing bijection on ideals with $\text{ann}(A \cap B) = \text{ann } A + \text{ann } B$ (Macaulay duality).

3. Proof of Theorem A

($\supseteq$) Parametrize V_J by $(x, -x, y, -y, z, -z)$. For odd k , the terms of e_k using a full $\pm$ pair carry the factor $x \cdot (-x)$ against a complement whose signed sum vanishes; terms taking exactly one element from each selected pair cancel under flipping one selected pair's signs. Explicitly e_1, e_3, e_5 all restrict to 0 on every V_J .

(Set-theoretic equality) If $e_1 = e_3 = e_5 = 0$ at a point s over the algebraic closure, then $\Pi(t - s_1) = t^6 + e_2 t^4 + e_4 t^2 + e_6$ is an even polynomial; since $\text{char} \neq 2$ its root multiset is invariant under $t \mapsto -t$ and the multiplicity of the root 0 is even. Hence $\{s_0, \dots, s_5\} = \{\alpha, -\alpha, \beta, -\beta, \gamma, -\gamma\}$, and some matching of the indices pairs each value with an opposite one: $s \in V_J$ for some J . Thus $V(e_1, e_3, e_5) = \cup_J V_J$.

(Ideal equality) The variety of the 3-generated ideal (e_1, e_3, e_5) has dimension 3, so the ideal has height 3 and its generators form a regular sequence; the quotient is a complete intersection, Cohen–Macaulay and unmixed, of Bézout degree 15. Its support consists of fifteen pairwise-distinct linear planes of degree 1; unmixedness forces every component to appear, and the degree count forces every multiplicity to equal 1. Generically reduced plus (S1) gives reduced, so the ideal is radical and equals $I(\cup_J V_J) = \cap_J I_J$. Finally e_2 restricts to $-(x^2 + y^2 + z^2) \neq 0$ on every V_J , so the odd ideal is exact. $\square$

4. Proof of Theorem B

Lemma 1 (digit transport across frames). Between any two frames the digit vectors of an element are related by fixed S -linear matrices: writing each reduced monomial μ of the first frame in the second frame's basis, $\mu = \sum_\rho F(a_{\mu\rho})\rho$ with $a_{\mu\rho} \in S$, additivity and multiplicativity of F give $t_\rho^{(y)} = \sum_\mu a_{\mu\rho} t_\mu$, and symmetrically with the inverse matrix. Proof. Substitute and use uniqueness of the digit expansion (K). $\square$

Lemma 2 (relative digit descent). Fix a matching J and a frame y whose first three coordinates are the leg forms. Then $I_J + m_Q = (y_1, y_2, y_3, y_4^Q, y_5^Q, y_6^Q)$ is a monomial ideal of the frame (by L2), and $t \in I_J + m_Q$ if and only if every J -frame digit $t_\rho^{(J)} \in I_J + m_{Q'}$. Proof. Membership in a monomial ideal is monomial-by-monomial, and the y -digit classes partition monomials by exponents mod 3. For reduced ρ , a monomial of $F(t_\rho)\rho$ is divisible by y_i ($i \leq 3$) iff the corresponding monomial of t_ρ is; and divisible by y_j^Q ($j \geq 4$) iff $3e + \rho_j \geq Q$ with $\rho_j \leq 2$ and $Q = 3Q'$, iff $e \geq Q' - \rho_j$ — the threshold is exact: $e = Q' - 1$ gives $3Q' - 3 + \rho_j < Q$, while $e = Q'$ gives $\geq Q$. $\square$

Lemma 3 (frame independence). Since $I_J + m_Q$ is an ideal and, by Lemma 1, the standard digits are S -linear combinations of the J -frame digits (and conversely), the conditions $[\forall \rho:$

$t_{\rho}^{(J)} \in I_J + m_{Q'}$ and $[\forall \mu: t_{\mu} \in I_J + m_{Q'}]$ are equivalent. $\square$

Inductive step. Let $t \in \cap_J (I_J + m_{Q'})$. For each J , Lemmas 2 and 3 place the standard digits t_{μ} in $I_J + m_{Q'}$; as this holds for every J simultaneously, $t_{\mu} \in \cap_J (I_J + m_{Q'}) = E + m_{Q'}$ by the inductive hypothesis — note the hypothesis is the full fifteen-fold intersection, applied whole, at every rung. Write $t_{\mu} = \varepsilon_{\mu} + w_{\mu}$ with $\varepsilon_{\mu} \in E$, $w_{\mu} \in m_{Q'}$, and reassemble: $t = \sum F(\varepsilon_{\mu})\mu + \sum F(w_{\mu})\mu$. Every ideal contains the cubes of its elements, so $F(\varepsilon_{\mu}) \in E$ and the first sum lies in E ; and $w \in m_{Q'}$ implies $w^3 \in m_Q$ (freshman's dream on $\Sigma h_{11}^{Q'}$), so the second lies in m_Q . Hence $t \in E + m_Q$. The same argument proves the general Descent Principle verbatim. $\square$

Base case ($v = 1$, $Q = 3$). In degrees 0–2 the ideal m_3 vanishes and both sides equal E_d by Theorem A. In degrees ≥ 13 every monomial in six variables has an exponent ≥ 3 and both sides fill S_d . In degrees ≥ 7 the left side fills S_d because each quotient $S/(I_J + m_3) \square F_3[x,y,z]/(x^3,y^3,z^3)$ has top degree 6. The remaining finite statement — equality in degrees 3–12 — is verified exactly by two independent computations: primal (exact ranks over F_3 of both sides; degrees 3–8: 27, 86, 216, 447, 792, 1287 on each side) and dual (Gorenstein duality identifies the assertion with $(m_3:e_1) \cap (m_3:e_3) \cap (m_3:e_5) = \sum_J (\pi_J^2) + m_3$, π_J the product of J 's three forms, since $\text{ann}(\bar{I}_J) = (\pi_J^{Q-1})$ is principal in the leg frame; measured equal at every degree 0–11, the values 0,0,0,6,36,126,336,702,1237,1981,2997,4367, with degree 12 by inspection of the socle monomial). Consistency: $\dim S/(E+m_3)$ has Hilbert function 1,5,15,29,40,36,15, total 141 = $|X_1(F_3)|$. Scripts and logs are published with the corpus. $\square$

5. Application: the Fermat tower and Conjecture 1.2

Theorem B is the campaign statement ($\square_{34}$): $\cap_J (I_J + m_v) = (\cap_J I_J) + m_v$ at every level. The campaign's sealed corpus establishes (with sufficiency direction, i.e. the direction used here): ($\square_{34}$) implies $\dim T(v) \geq P(3^v)$, where $\dim T(v) = 3^{6v} - \dim \cap_J \bar{I}_J$ and $P(3^v) = |X_v(F_3)|$ is the point count of the union of the leg planes, proven for all v by inclusion-exclusion (anchors 141, 7761, 263901); the reverse inequality $\dim \leq \text{points}$ is unconditional; hence $h(v) = 0$ and the rank chain $P(3^v) \geq \text{rank}_{F_3}(\Delta_v) \geq \dim T(v)$ collapses to equality, giving $\text{rank}_{F_3}(\Delta_v) = \text{rank}_Q(\Delta_v) - \text{drop zero} - \text{for every } v$; finally $\text{drop}(\Delta_v) = 0$ for all v is equivalent to $\text{Tors}_3(\text{coker } \Delta_v) = 0$ for all v , the saturation of $\Sigma 1_J = \text{im}(\Delta)$ at every level of the tower, which is Conjecture 1.2 of Degtyarev-Shimada in dimension 4. The corpus addresses (sections 73.3, 239, 242, 244, 245 of the campaign master, and the Nail theorem's bridge) are cited in the companion document; independent re-derivation of the chain is invited as part of record discipline.

6. Remarks

(i) Both theorems are S_6 -equivariant; the fifteen matchings form one orbit and Theorem A presents the ideal of the orbit of planes as a complete intersection on the odd half of the fundamental invariants, with Bézout number equal to the orbit size. (ii) In characteristic 3 the elementary symmetric polynomials commute with Frobenius, $e_k(s)^3 = e_k(s^3)$, and the dual generators scale literally: $\pi_J^{Q-1} = F(\pi_J^{Q'-1}) \cdot \pi_J^2$ — the exact self-similarity that the descent converts into a proof. (iii) The Descent Principle is stated for arbitrary linear arrangements; the authors are not aware of a prior appearance in this form and would welcome references.

References

- [1] A. Degtyarev, I. Shimada. On the topology of projective subspaces in complex Fermat varieties. *J. Math. Soc. Japan* 68:3 (2016), 975–996. [arXiv:1405.4683](#).
- [2] E. Aljovin, H. Movasati, R. Villaflor. Integral Hodge conjecture for Fermat varieties. *J. Symbolic Computation* 95 (2019), 177–184. [arXiv:1711.02628](#).
- [3] T. Shioda. The Hodge conjecture for Fermat varieties. *Math. Ann.* 245 (1979).
- [4] E. Kunz. Characterizations of regular local rings of characteristic p . *Amer. J. Math.* 91 (1969).
- [5] Operación Glotón corpus: theorems, gates and logs. github.com/tretoef-estrella (campaign master sections cited in text; verification scripts PROBE_V57/V58/V59).

Written in pencil where pencil suffices; the finite base is verified twice by independent exact computation, and every number is published. The authors invite the community to attack the base scripts and the cited chain — records are records only after they survive other hammers.